

Lesson 12: Bayesian Inference: One Dry Day; Three Ways to Answer — Only One Keeps the Doubt

This lesson covers the next two topics from the PDF:

1. **Bayesian Inference: One Dry Day**
2. **Three Ways to Answer — Only One Keeps the Doubt**

The central lesson is that one observation should update our uncertainty, not erase it. A point estimate can be useful, but it hides how uncertain we remain. Bayesian inference retains a posterior distribution over plausible parameter values, allowing us to distinguish a best guess from the complete state of updated belief.

1. Bayesian Inference: One Dry Day

1.1 The unknown quantity

Let q be the probability that it rains on any particular day. The parameter satisfies

$$0 \leq q \leq 1.$$

We observe one day and find that it was dry. Let

$$x = 0$$

denote no rain. Under the Bernoulli model,

$$P(X = 0 \mid q) = 1 - q.$$

The question is not “what was the probability of this observation after it happened?” The observation is already known. The inferential question is:

After observing one dry day, what should we believe about the unknown rain probability q ?

1.2 Step 1: prior

The PDF chooses a uniform prior:

$$p(q) = 1, \quad 0 \leq q \leq 1.$$

This prior gives equal density to all rain probabilities between 0 and 1 and zero density outside that interval. It represents a deliberately simple starting point, not evidence that all values are equally realistic in the physical world.

The prior is a probability density, so

$$\int_0^1 p(q) dq = \int_0^1 1 dq = 1.$$

1.3 Step 2: likelihood

For a dry day,

$$L(q) = P(X = 0 \mid q) = 1 - q.$$

The likelihood is a score for each candidate value of q :

Candidate q	Likelihood of a dry day $1 - q$	Interpretation
0.1	0.9	A dry day is easy to explain.
0.5	0.5	A dry day is neither surprising nor especially expected.
0.9	0.1	A dry day is relatively difficult to explain.

The likelihood is not a probability distribution over the candidate values of q . In particular,

$$\int_0^1 (1 - q) dq = \frac{1}{2} \neq 1.$$

It becomes part of a posterior distribution only after multiplication by the prior and normalization.

1.4 Step 3: posterior

Bayes Theorem gives

$$p(q \mid x = 0) = \frac{P(x = 0 \mid q)p(q)}{P(x = 0)}.$$

The numerator is

$$P(x = 0 \mid q)p(q) = (1 - q) \cdot 1 = 1 - q.$$

The evidence is

$$\begin{aligned} P(x = 0) &= \int_0^1 P(x = 0 \mid q)p(q) dq \\ &= \int_0^1 (1 - q) dq \\ &= \frac{1}{2}. \end{aligned}$$

Therefore, the normalized posterior density is

$$p(q \mid x = 0) = 2(1 - q), \quad 0 \leq q \leq 1.$$

The posterior integrates to one:

$$\int_0^1 2(1 - q) dq = 1.$$

The posterior decreases as q increases. Small values of q are more supported after a dry day, but larger values are not automatically impossible.

1.5 Posterior summaries

The posterior provides several possible summaries.

Posterior mode / MAP

The posterior density $2(1 - q)$ is largest at

$$\hat{q}_{\text{MAP}} = 0.$$

This is the most probable point in the sense of the posterior density's mode. For a continuous parameter, however, every exact point has probability zero, so “most probable point” should be understood as “point at which the density is highest,” not as an event with positive probability.

Posterior mean

The posterior mean is

$$\begin{aligned}\mathbb{E}[q \mid x = 0] &= \int_0^1 q \, 2(1 - q) \, dq \\ &= \frac{1}{3}.\end{aligned}$$

This is different from the MAP value 0. The mean averages over all posterior possibilities; the mode identifies the peak.

Posterior probability of a range

For any $0 \leq a \leq b \leq 1$,

$$P(a \leq q \leq b \mid x = 0) = \int_a^b 2(1 - q) \, dq.$$

For example,

$$\begin{aligned}P(q \leq 0.5 \mid x = 0) &= \int_0^{0.5} 2(1 - q) \, dq \\ &= 0.75.\end{aligned}$$

Thus, after one dry day, the posterior probability that the rain chance is at most 0.5 is 75% under this model and prior.

The posterior probability that $q > 0.6$ is

$$\begin{aligned}P(q > 0.6 \mid x = 0) &= \int_{0.6}^1 2(1 - q) \, dq \\ &= (1 - 0.6)^2 \\ &= 0.16.\end{aligned}$$

This makes the uncertainty explicit: $q = 0.6$ is disfavoured, but a relatively high rain probability has not been ruled out by one dry day.

1.6 Repeated dry days

If we observe d dry days and no rainy days, then

$$P(D \mid q) = (1 - q)^d.$$

With the uniform Beta(1, 1) prior,

$$q \mid D \sim \text{Beta}(1, d + 1).$$

Its density is

$$p(q \mid D) = (d + 1)(1 - q)^d.$$

As d grows, the posterior concentrates more strongly near smaller values of q . Even then, the conclusion depends on the model: the days must be reasonably comparable, the rain probability must be treated as stable, and the observations must satisfy the assumed dependence structure.

2. Three Ways to Answer — Only One Keeps the Doubt

The PDF contrasts three ways to summarize what we know after data are observed.

2.1 Answer 1: a single point estimate

One could report a single value such as

$$q = 0,$$

because the posterior mode is 0. This is the MAP answer for the one-dry-day example.

A point estimate is concise and may be suitable for a decision requiring one number. However, it hides the fact that the posterior assigns positive density across the whole interval. Reporting only 0 may be misread as proof that rain is impossible.

The ML estimate also gives a point. With one dry observation and the Bernoulli likelihood,

$$\hat{q}_{\text{ML}} = 0.$$

This is an extreme estimate because the sample contains no rainy days. It does not itself communicate how little data produced that result.

2.2 Answer 2: an interval or probability statement

A second approach is to report an interval or a posterior probability, such as

$$P(q \leq 0.5 \mid x = 0) = 0.75,$$

or a credible interval containing a specified posterior probability.

This keeps more uncertainty than a point estimate. It communicates that a range of values remains plausible. The interval is still a summary, however, and different intervals can omit different features of the posterior.

A Bayesian credible interval is interpreted using the posterior distribution and the model/prior assumptions. For example, a 95% credible interval is a set with posterior probability 0.95, not a statement about the long-run coverage of a repeated-sampling procedure.

2.3 Answer 3: the full posterior distribution

The third approach is to report the entire posterior:

$$p(q \mid x = 0) = 2(1 - q), \quad 0 \leq q \leq 1.$$

This is the answer that “keeps the doubt.” It retains information about the relative support for every possible q , allowing the reader to calculate the mode, mean, variance, quantiles, interval probabilities, and predictions.

The full posterior tells us:

1. small values of q are more supported than large values;
2. the data did not force one exact value;
3. uncertainty remains substantial after only one observation;
4. different summaries, such as the mode and mean, answer different questions.

2.4 Comparing the three answers

Answer type	Example here	What it preserves	What it hides
Point estimate	MAP $q = 0$	A single preferred value.	Spread, skewness, and remaining plausible values.
Interval/probability statement	$P(q \leq 0.5 \mid x = 0) = 0.75$	Some measure of uncertainty.	Details outside the selected interval or event.

Full posterior	$p(q \mid x = 0) = 2(1 - q)$	All posterior uncertainty and shape.	It may be less convenient than one number for immediate decisions.
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The full posterior is not always required in communication, but it is the fundamental Bayesian object from which the other summaries can be derived.

2.5 Posterior predictive probability

The posterior can also answer a future question. Suppose tomorrow's rain indicator is X_{new} , with

$$P(X_{\text{new}} = 1 \mid q) = q.$$

The posterior predictive probability of rain tomorrow is

$$\begin{aligned} P(X_{\text{new}} = 1 \mid x = 0) &= \int_0^1 P(X_{\text{new}} = 1 \mid q) p(q \mid x = 0) dq \\ &= \int_0^1 q \cdot 2(1 - q) dq \\ &= \frac{1}{3}. \end{aligned}$$

In this particular model, the posterior predictive probability equals the posterior mean of q . It is not equal to the MAP value 0 because the MAP is a point summary, whereas prediction averages over parameter uncertainty.

2.6 What the example does not claim

The example does not establish that the actual probability of rain is zero, one-third, or any other objectively known number. Those results are conditional on the Bernoulli model, the stability assumption for q , the uniform prior, and the observation that one day was dry.

If weather changes with season, location, atmospheric conditions, or other covariates, a constant single-parameter Bernoulli model may be inadequate. Bayesian inference preserves uncertainty within the model; it does not automatically correct a poor model.

3. Complete mental model

The PDF's three-step Bayesian update is

$$\text{prior } p(q) \rightarrow \text{likelihood } P(x | q) \rightarrow \text{posterior } p(q | x)$$

For one dry day,

$$1 \times (1 - q) \rightarrow 2(1 - q).$$

The prior starts the analysis, the likelihood scores candidate values using the data, and the posterior reweights the prior. The best single answer may be useful, but the full posterior is the answer that retains the remaining doubt.

4. Check your understanding

1. Why is $P(X = 0 | q) = 1 - q$ for a dry day?
2. Why is the likelihood $1 - q$ not yet a probability distribution over q ?
3. Derive the evidence for one dry day under a $\text{Uniform}(0, 1)$ prior.
4. Derive the posterior density $p(q | x = 0) = 2(1 - q)$.
5. Why is the MAP estimate 0 but the posterior mean $1/3$?
6. Calculate $P(q \leq 0.5 | x = 0)$.
7. Calculate $P(q > 0.6 | x = 0)$.
8. Explain what information is lost when only the MAP estimate is reported.
9. Derive the posterior predictive probability of rain tomorrow.
10. Name the modelling assumptions under which the one-dry-day conclusion is valid.

References

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- [3] OpenStax, Principles of Data Science, 3.4 Probability Theory
- [4] Stanford University, CS229 Lecture Notes